# DepthWizard: Finding Building and Ground Heights from One Satellite Photo for Disaster Response

HS202: Human Geography · Project Report · Group 29


| | |
|---|---|
| Himanshu Patel | 2024AIB1007 |
| Shlok Vaidya | 2024AIB1016 |
| Agampreet Singh | 2024CSB1097 |
| Saaransh Garg | 2024AIB1015 |
| Takshita Nagrale | 2024AIB1018 |
| Saloni Mahajan | 2024CSB1149 |
| Peruru Smitha Reddy | 2024CSB1141 |
| Akshita | 2024CSB1098 |
| Aryan Goyal | 2024CSB1102 |
| Arahana Garg | 2024MCB1261 |

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Code and demo: https://github.com/hima1323/terravista-depthwizard


**Abstract**

When a flood or landslide strikes, rescue teams need to know how tall buildings are, how high the ground lies and where water will reach. Most of India has no recent 3D map that can answer this. DepthWizard estimates the height of every building, tree and patch of ground from one ordinary satellite photograph, ties those heights to sea level, and says how accurate they are. Our main new idea is a height model built on Mamba, a recent kind of neural network that reads an image in one sweep. It is nine times smaller and 5.2 times faster than the published model it learns from, although its first version is less accurate. Around the model we add a ground-level correction that uses NASA's satellite lasers, an error range for every building, and eighteen tools for disaster teams. With the published model, the average height error is 3.90 m on 150 aerial test tiles, where a flat-ground guess is off by 5.65 m, and 4.85 m at eight Indian sites. On the 2023 Teesta flood in Sikkim, free satellite images map 1.28 $km^2$ of damaged valley floor.



**Keywords:** height from a single image; Mamba; satellite lasers; flood mapping; disaster management; India.


## 3. Definition of the Problem

### 3.1 Problem statement

Estimate a reliable height, in metres, for every point of a single satellite photograph, and show how far that height can be trusted.

### 3.2 Origin/identification of the problem

In a flood, height decides the outcome: whether water enters a house, which roofs can shelter people, which slopes may slide. Yet planners rarely have height data. Aircraft laser surveys are accurate but costly, and few Indian districts have one. Two-angle satellite pairs give height, but

a pair seldom exists for the place and date of an emergency, and free worldwide maps record one value every 30 m, too coarse to see a building. A single satellite photograph, though, exists for almost every place and date. The Teesta flood of October 2023, studied in Section 11.5, is an example: no public map of that valley showed single buildings.

### 3.3 Methodology adopted to identify the problem

We did not run surveys, key-informant interviews or focus-group discussions. We identified the problem in three ways. We read the published work on height from single images, on free ground maps and on satellite lasers. We discussed within the team what a disaster manager would need from a height map. And before building anything, we measured how far the free ground map sits from laser-measured ground at Indian sites. In towns it was 5.28 m off on average, which told us where to work first.

### 3.4 Detailed description of the identified problem

Height from one image is hard to use for three reasons. First, the image is ambiguous: seen from above, a tall building and a short one can look alike, so models guess from shadows and roof size and tend to make tall objects too short. Second, flood work needs height above sea level, while a model gives height above the ground; in towns and forests the free ground map includes roofs and tree tops, so its ground sits too high. Third, most tools give a height map without any error figure, and a relief officer cannot tell whether a 2 m flood depth is good to half a metre or to five. The tool must also work in cities, villages, hills and forests; on a hillside the slope itself can be mistaken for height.

### 3.5 Current developments in the domain

Neural networks can now estimate depth from one photograph [6]. RS3DAda [3] adapted this to satellite images by training on computer-generated cities, and it is the model we start from. Depth Anything V2 [7] is a strong general model trained on ordinary photographs. Mamba [1] and MambaVision [2] are newer network designs that cost less to run. For the ground, Copernicus GLO-30 [9] is a free worldwide map and FABDEM [10] is a version with buildings and forest removed. NASA's ICESat-2 [11, 12] and GEDI [13] satellites measure heights with lasers along narrow tracks, and Song and others [8] recently showed that such laser points can correct single-image heights. Meta has published a worldwide map of tree heights [14]. India has its own Cartosat images and CartoDEM ground map from NRSC, but both need a login, so we have not tested them yet.

### 3.6 Need and significance of resolving the problem

A trusted height map would let a district office see before the monsoon which streets a water rise would reach, route rescue teams away from deep water, choose helipad and camp sites, and count damaged buildings afterwards. The error figure matters as much: a planner who knows a map is good to 3 m in towns and poor under forest can use it with care. Table 1 shows what each existing source lacks.

**Table 1:** Ways of getting heights for disaster work.

| Approach | Needs | Detail | Sea level | Error stated |
|---|---|---|---|---|
| Aircraft laser survey | A survey flight | Very fine | Yes | Yes |
| Two-angle satellite pair | Two matched images | Fine | Yes | Partly |
| Free ground maps [9, 10] | Nothing | 30 m | Yes | No |
| RS3DAda [3] | One image | Pixel | No | No |
| DepthWizard (ours) | One image and free data | Pixel | Yes | Yes |

## 4. Aims/Objectives/Goals

Our aim is one offline tool that turns a single satellite photograph into a height map that disaster teams can use and check. Each objective in Table 2 closes one gap from Section 3 and has a test.

**Table 2:** Objectives and how each is measured.

| # | Objective | How success is measured |
|---|---|---|
| O1 | A height for every point of one image | Smaller error than a flat-ground guess on unseen data |
| O2 | Heights above sea level | Closer to laser-measured ground than the free ground map |
| O3 | An error range on each building | The range holds 90% of the time in unseen cities |
| O4 | Checking inside the app | Error figures for each scene and type of land |
| O5 | Disaster tools that are easy to find | Every tool opens from one search box |
| O6 | Works offline on a laptop | Measured time for one full scene |
| O7 | A cheaper height model (Mamba) | Speed, size and accuracy on the same tests as RS3DAda |

## 5. Tools and Techniques (Interventions)

### 5.1 Height model

RS3DAda [3] is a DINOv2 vision transformer [4] with two DPT decoder heads [6]. For each pixel it returns a height above the ground, $h$, and scores $z$ for eight land-cover classes such as building, tree, road and water. We chose it because it was trained for satellite images, so it predicts height above the ground and not distance from a camera. It also gives land cover in the same pass, which the ground step below needs, and its weights are public, so anyone can repeat our numbers. We did not retrain it.

A scene is far larger than the network's 518-pixel input, so we cut it into tiles that overlap by half. Each tile $k$ is read four times, once as it is and once for each flip, and the four answers are averaged. The tiles are then blended with weights $w_k(x)$ that are 1 in the middle of a tile and fall linearly to near 0 at its edge:

$$\hat{h}(x) = \frac{\sum_k w_k(x)\,\bar{h}_k(x)}{\sum_k w_k(x)}, \qquad \bar{h}_k = \tfrac{1}{4}\sum_{f=1}^{4} h_k^{(f)}, \qquad u_k = \mathrm{std}_f\ h_k^{(f)}. \tag{1}$$

The spread $u$ of the four readings, blended the same way, shows where the model is unsure, and Section 5.4 turns it into an error range.

### 5.2 From height above ground to height above sea level

The model gives the height $h$ of each point $x$ above the ground under it. Adding the height of the ground gives the height above sea level:

$$\mathrm{DSM}(x) \;=\; \mathrm{DTM}(x) \;+\; h(x). \tag{2}$$

The ground map starts as GLO-30 [9], a 30 m surface model that sits on rooftops and tree tops in towns and forests. For every 30 m cell $c$ we average two things from our own outputs: the mean height above ground $m_c$, and the open-ground fraction $g_c$, the share of the cell that the land-cover map calls road, bare soil or water. On flat scenes, with less than 50 m of relief, the cleaned cell height is

$$\mathrm{DTM}_c = \mathrm{GLO}_c - m_c, \qquad \text{trusted with weight } g_c^2, \tag{3}$$

and cells whose weight is below 0.25, mostly roofs or trees, are filled in from their neighbors. In hills the slope itself must survive, so every cell keeps its shape and loses only a share $\alpha$ of the object height, $\mathrm{DTM}_c = \mathrm{GLO}_c - \alpha\, m_c$, with weight $0.2 + 0.8\, g_c^2$.

We then level the whole map with ICESat-2 photons classed as ground [12]. For ground photons $j$ at heights $z_j$, the shift is the median difference

$$\delta = \operatorname*{median}_j \big(z_j - \mathrm{DTM}(x_j)\big), \qquad \mathrm{DTM} \leftarrow \mathrm{DTM} + \delta. \tag{4}$$

The median keeps a few bad photons from pulling the result. Two rules came from real failures. First, the shift is applied only with at least 50 ground photons from at least two passes; one pass is a single line and can carry a local error into the whole scene. Second, a ground photon counts as a roof, and is dropped, when it falls on a building pixel grown by 2 m and also stands more than 3 m above the median of the other ground photons. Either sign alone was not enough in our tests.

A user with $K$ surveyed points of known height $z_k$ at positions $(x_k, y_k)$ can correct the map further. We find the level $a$, scale $b$ and tilt $c, d$ that best fit the map values $\hat{z}_k$:

$$\min_{a,b,c,d} \sum_{k=1}^{K} \big(z_k - a - b\,\hat{z}_k - c\,x_k - d\,y_k\big)^2. \tag{5}$$

One point fixes only $a$, two or three also fix $b$, and four or more fix the tilt as well. Each fit reports its error on a left-out point. This also lets a plain photograph without location data reach heights above sea level.

### 5.3 Tree heights

The model sees trees at only 18 to 60% of their true height. Tree pixels can take their height from Meta's canopy map [14], which is resampled onto the image grid. The other route is a single scale for the tree class, fitted from GEDI laser shots [13] that fall on trees, each covering a circle of 12.5 m radius:

$$s_{\mathrm{tree}} = \operatorname*{median}_k \frac{H_k}{\hat{h}_k}, \qquad \hat{h} \leftarrow s_{\mathrm{tree}}\,\hat{h} \ \text{ on tree pixels}, \tag{6}$$

where $H_k$ is the laser's canopy height, its 98th-percentile return, and $\hat{h}_k$ is the model's height over the same footprint.

### 5.4 An error range for every building

The spread $u$ ranks errors on flat ground well, with a rank correlation of 0.44, but says almost nothing on tall objects, where it is $-0.03$. So we do not show it as an error bar. We use split conformal prediction [18] instead. On $n$ calibration buildings with known errors $e_i$ we compute scores and a range

$$r_i = \frac{|e_i|}{u_i + u_0}, \qquad \hat{q} = r_{(\lceil (n+1)(1-\alpha) \rceil)}, \qquad C(x) = \big[\,\hat{h}(x) \pm \hat{q}\,\big(u(x) + u_0\big)\big], \tag{7}$$

where $r_{(k)}$ is the $k$-th smallest score, $\alpha = 0.1$ and $u_0 = 0.25$ m. If new buildings behave like the calibration ones, conformal theory guarantees $P\big(h \in C(x)\big) \geq 1 - \alpha$, nine times in ten. A new city may break that condition, so we test the range on cities left out of the calibration.

### 5.5 Disaster models

A simple flood marks as wet every cell below a chosen level that connects to the river or the scene edge. For water that moves over time we solve the local-inertial shallow-water equations of Bates

and others [15] on the height grid. The flow $q$ between two neighboring cells, per metre of their shared edge, is updated as

$$q_{t+\Delta t} \;=\; \frac{q_t - g\, h_f\, \Delta t\, \partial \eta / \partial x}{1 + g\, \Delta t\, n^2\, |q_t| \,/\, h_f^{7/3}}, \qquad h_{t+\Delta t} = h_t + \frac{\Delta t}{\Delta x} \sum_{\text{edges}} \pm q, \tag{8}$$

where $\eta$ is the water surface, $h_f$ the depth through which water can flow, $g$ gravity and $n$ Manning's roughness. We set $n$ from the land cover, from 0.016 for roads to 0.1 for trees. The time step follows the stability limit $\Delta t = \min\!\left(5\,\text{s},\; 0.7\,\Delta x / \sqrt{g\, h_{\max}}\right)$. On steep urban ground the scheme can blow up, so we cap the flow at the critical value $|q| \le h_f \sqrt{g\, h_f}$, a Froude number of 1, and limit each cell's outflow to the water it holds. With these limits, water in equals water stored plus water out to within 5% in our tests.

A real flood can be loaded as a map, or taken from a Sentinel-1 radar image in decibels. Water is dark in radar, so the app picks the threshold $\tau$ with Otsu's method [16], the value that best separates the two classes:

$$\tau^{*} = \arg\max_{\tau} \; \omega_0(\tau)\, \omega_1(\tau)\, \big(\mu_0(\tau) - \mu_1(\tau)\big)^2, \tag{9}$$

where $\omega$ and $\mu$ are the share and mean of the pixels on each side of $\tau$. A predicted flood is then scored against the real one with the critical success index

$$\text{CSI} \;=\; \frac{TP}{TP + FP + FN}, \tag{10}$$

where $TP$ counts cells both maps call wet, and $FP$ and $FN$ count cells only one of them calls wet. Unlike plain accuracy, it ignores cells dry in both maps.

Evacuation routes are shortest paths found with Dijkstra's method [19] over a cost grid. A cell's cost is

$$c = c_{\text{lc}} \Big(1 + \frac{s}{s_0}\Big) \Big(1 + 3\,\frac{d}{d_{\max}}\Big), \tag{11}$$

where $c_{\text{lc}}$ depends on the land cover, $s$ is the slope and $d$ the water depth. Buildings, slopes above a limit, and water deeper than $d_{\max}$ cost infinity, so routes never cross them. Landslide risk uses the slope classes of the Indian Standard IS 14496 [17]. The number of people at risk is the floor area of each affected building, times its number of floors and the share that is lived in, divided by the floor space one person needs. The app labels all of these outputs as scenarios, not forecasts.

### 5.6 The application

DepthWizard runs on an ordinary laptop and works without internet once a scene has been opened. It has a scene library and six working pages that follow a disaster from start to finish, shown in Figure 2. Table 3 lists its eighteen tools, which switch on in place, and any of them can be found by typing its name in a search box. The 3D view can be circled, flown through freely or walked at street level, and a flight along a drawn path can be recorded as a video for briefings. Results can be saved as standard map and 3D files that open in QGIS, ArcGIS or Bhuvan.

**Table 3:** The eighteen tools and the question each answers.

| Tool | Question it answers |
|---|---|
| *Explore* | |
| Line of sight | What can be seen from this mast or tower? |
| Volume and area | How much debris is in this pile? |
| Drone flight plan | Which route keeps a drone above every roof? |
| Rooftop solar | How much power could each roof produce? |
| Map overlay | Where are the streets on the 3D view? |
| Mosaic | How do neighbouring scenes join up? |
| *Validate and calibrate* | |
| Heights from shadows | Do shadow lengths confirm the heights? |
| Tree and tall-object fix | Can weak heights be corrected from other sources? |
| Error bars | How far could each height be off? |
| Terrain source | Can I use my own ground map? |
| GPS and field points | Can points measured on the ground correct the map? |
| *Prepare, Respond, Relief, Assess* | |
| Landslide risk | Which slopes may fail in heavy rain? |
| People at risk | How many people live in the affected buildings? |
| Observed flood | How well does the predicted flood match a real one? |
| Flowing water | Where does the water go, hour by hour? |
| Evacuation plan | Which camp should each family go to? |
| Damage before/after | Which buildings collapsed? |
| Situation report | What does a decision maker need on one page? |

## 6. Detailed Work Plan / Technological Interventions

Figure 1 shows the seven steps a scene goes through. A PNG or JPEG gives heights above the ground; a GeoTIFF, which carries its location, continues to heights above sea level. Table 4 gives the order of work. Most of the effort went into the fifth step, because the free ground map is wrong exactly where people live. Automated tests click through every tool on scenes with known answers, 38 checks in the main interface and 36 in the older viewer, and all pass.

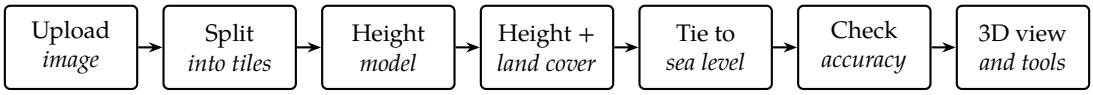


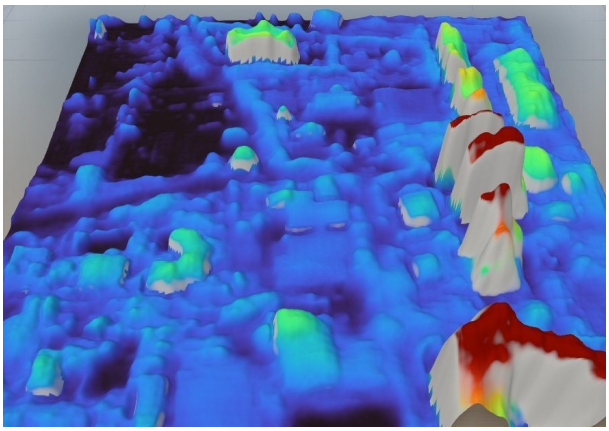

**Figure 1:** From one photograph to a checked 3D map.

**Table 4:** Work plan as carried out.

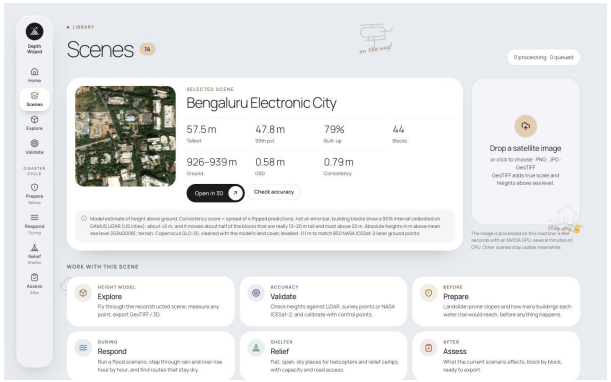
| Period | Work |
|---|---|
| Before 15 October | Study of the problem, choice of model, first height maps |
| 15 October | Checking inside the app, correction from surveyed points, installer, first disaster tools |
| 18 October | Test method with confidence ranges, register of every number, error range per building |
| 22 to 25 October | New interface, roof-point rule for lasers, training code for the Mamba model |
| 28 October to 5 November | Eighteen tools and the tool finder, tree-height correction, Teesta case study, Mamba training and model comparison |



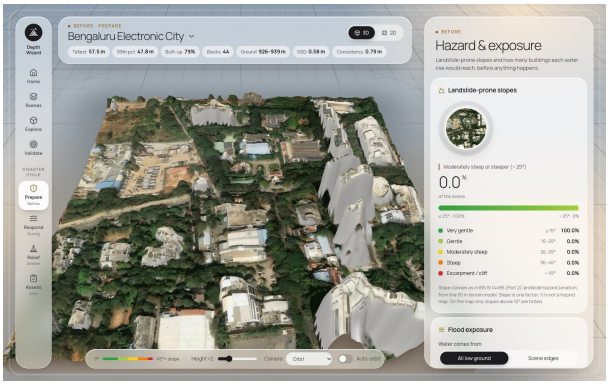
(a) 3D model with the photograph on it



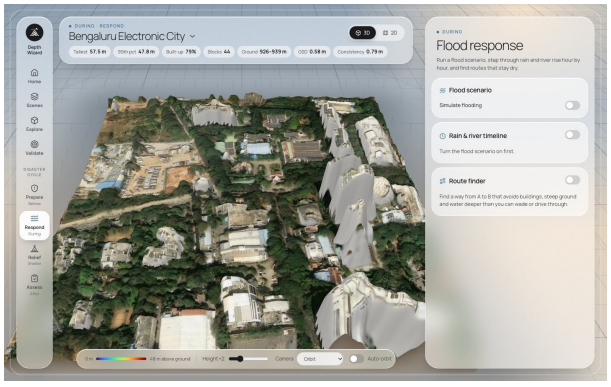
(b) Colored by height: 0 m blue, 48 m red



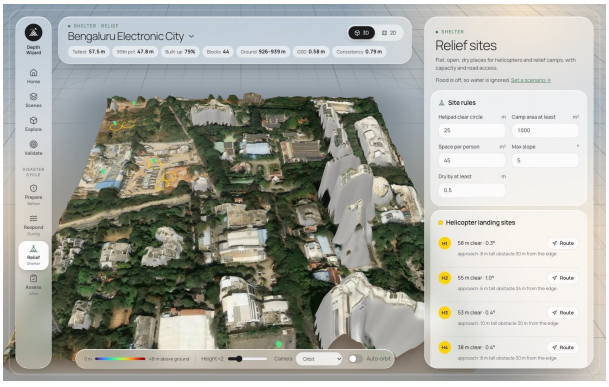

(c) Scene library and upload




(d) Prepare: risky slopes and water rise

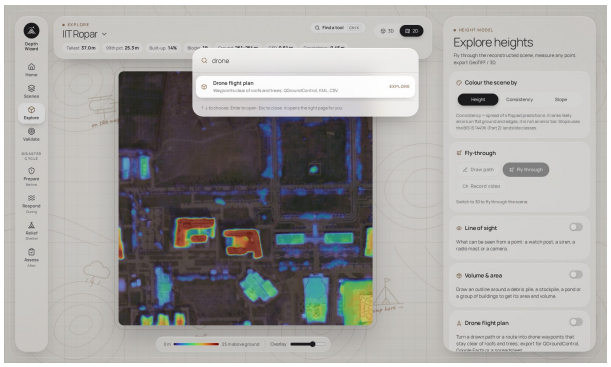



(e) Respond: flood scenario and routes

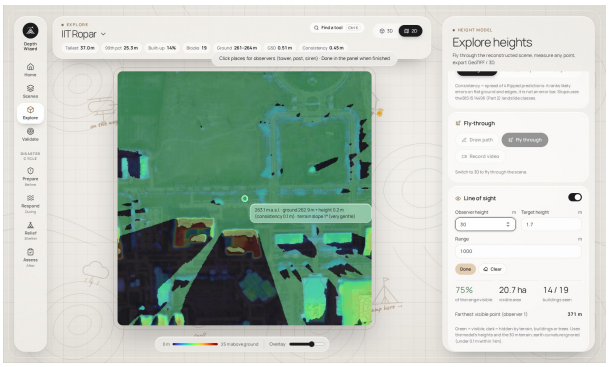



(f) Relief: helipad and camp sites

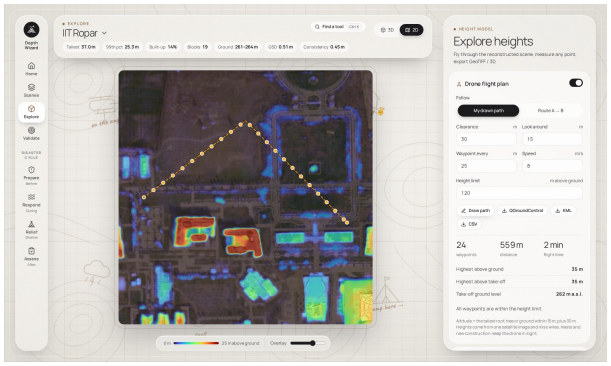
**Figure 2:** DepthWizard on a real scene, Bengaluru Electronic City, built from one satellite image. Heights in (a) and (b) are drawn at twice their true scale. Imagery: Esri World Imagery, for demonstration only.



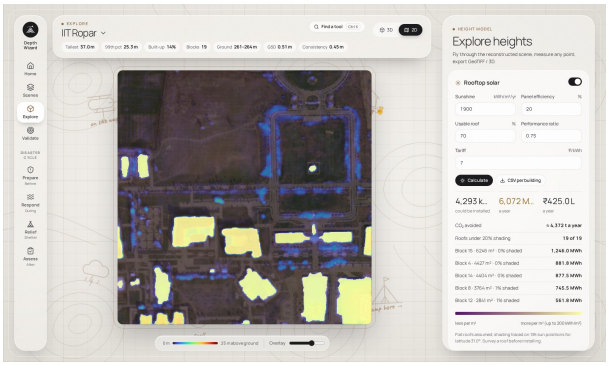
(a) Tool finder: typing “drone”



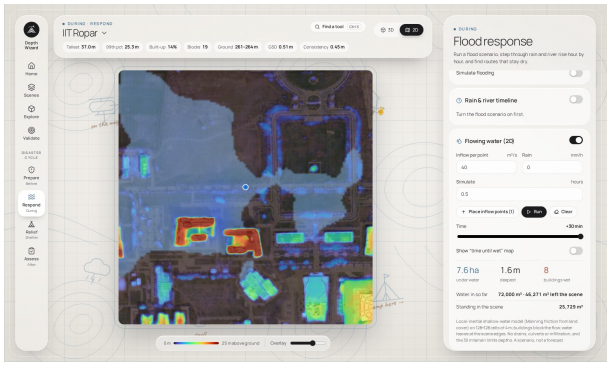
(b) Line of sight from a 30 m mast



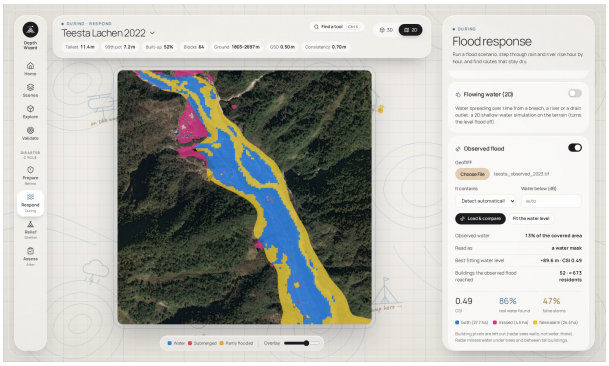
(c) Drone flight plan above every roof



(d) Rooftop solar with shading



(e) Flowing water from an inflow point



(f) Predicted against real flood, Teesta 2023

**Figure 3:** Tools running in the app. Panels (a) to (e): the IIT Ropar campus. Panel (f): the Teesta valley with the 2023 damage map.

## 7. Novelty / Innovation of the Proposed Interventions

### 7.1 A Mamba-based height model

RS3DAda reads an image with a transformer. The image is cut into $N$ patches, and every layer compares every patch with every other one through attention, which costs on the order of $N^2 d$ operations for patches described by $d$ numbers. Doubling the side of a tile quadruples $N$ and multiplies the attention cost by sixteen. That is why large satellite scenes are slow and need a strong graphics card.

Mamba [1] is a state-space model. It reads the patches one after another and keeps a running memory, the state $s_t$:

$$s_t = \bar{A}_t\, s_{t-1} + \bar{B}_t\, x_t, \qquad y_t = C_t\, s_t, \tag{12}$$

where $x_t$ is the $t$-th patch and $y_t$ the output. The matrices come from a continuous system through a step size $\Delta_t$, with $\bar{A}_t = \exp(\Delta_t A)$ and $\bar{B}_t \approx \Delta_t B_t$. What makes Mamba "selective" is that $\Delta_t$, $B_t$ and $C_t$ are computed from the input patch itself. The model can therefore decide, patch by patch, what to keep in memory and what to forget; it can keep a roof edge and pass over a plain field. One pass over $N$ patches costs on the order of $Nd$, which grows only in step with the image area.

MambaVision [2] adapts this to images in four stages. The first two are convolution blocks, which find edges and textures cheaply. The last two mix Mamba blocks with a few attention blocks, so the model can still relate distant parts of a scene, such as a building and the shadow it casts. The four stages produce feature maps at $1/4$, $1/8$, $1/16$ and $1/32$ of the input size, with 80, 160, 320 and 640 channels in the Tiny version we use. Because these already form a pyramid, each one goes through a single $1 \times 1$ convolution into RS3DAda's two DPT heads, which fuse the scales back to full resolution. Figure 4 shows the layout. The new model gives the same outputs as RS3DAda, so the rest of DepthWizard runs unchanged, and it has 40.7 million parameters against 366.5 million.

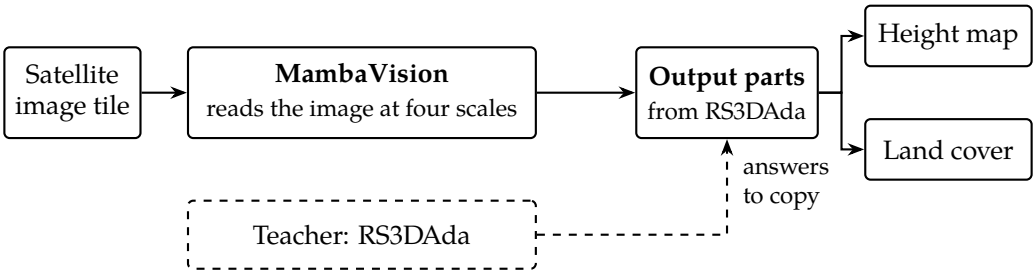



**Figure 4:** The Mamba-based height model and how it learns from RS3DAda.

Training a height model from nothing needs a very large labeled dataset. We used knowledge distillation [5] instead. The Mamba model, called the student and written S, looks at unlabeled images and learns to give the same answers as RS3DAda, the teacher, written T. Its training loss is

$$\begin{aligned}\mathcal{L} \;&= \ell\big(\hat{h}_{\mathrm{S}}, \hat{h}_{\mathrm{T}}\big) + \lambda\, \tau^2\, \mathrm{KL}\big(\sigma(z_{\mathrm{T}}/\tau) \,\big\|\, \sigma(z_{\mathrm{S}}/\tau)\big) \\ &\quad + \gamma\, \ell\big(\hat{h}_{\mathrm{S}}, h_{\mathrm{true}}\big),\end{aligned} \tag{13}$$

where $\ell$ is the smooth-L1 loss, $\sigma$ the softmax over land-cover classes and KL the Kullback–Leibler divergence. The first term pulls the student's heights towards the teacher's; smooth-L1 behaves like a squared error for small differences and like an absolute error for large ones, so a few wild pixels do not dominate. The second term matches the land-cover answers. Dividing the scores by a temperature $\tau = 2$ softens both distributions, so the student also learns which classes the teacher found close, and the factor $\tau^2$ keeps that term's gradients at the right size. The third term, used when labels exist, compares the student with true heights from the synthetic SynRS3D cities, which lets it learn things the teacher gets wrong. We used $\lambda = 0.5$, the AdamW optimizer, learning rates of $10^{-4}$ for the heads and $10^{-5}$ for the pretrained encoder with a 500-step warm-up and cosine

decay, and 20,000 steps of eight 512-pixel crops with random flips and turns. Section 11 reports how it did.

### 7.2 Other new parts

Most single-image tools stop at height above ground; we reach sea level with Eqs. (3) and (4). Their two safety rules came from mistakes we saw: in Silent Valley one laser track along a forested valley floor would have moved the whole scene by 46.59 m, and in Chandigarh all 35 "ground" laser points we held back for checking were on roofs. The app now refuses both cases. Each building also gets an error range whose reliability we measured in cities the method never saw. Every number in this report is linked in a register to the file that produced it.

## 8. Approaches that can be taken to Implement Intervention Plans

DepthWizard can reach users in five ways. District and state offices can install the offline app with one click; a full scene took 42 seconds on a laptop with 8 GB of memory. Its outputs are standard files, so NDMA and state authorities can load them into the software they already use. Field staff can record GPS points and known heights on a phone page that sends them back as check points. Anyone can try the hosted web demo on Replit, kept awake by a check every five minutes. Finally, the faster Mamba model on a central server could process many districts on one graphics card. Each rollout would start with a test on Indian imagery, then a pilot with one office through a monsoon, staff training, and feedback through the app's checking page.

## 9. Possible Constraints and Barriers to Implementation; Design Issues

Table 5 lists the limits we know of; the largest is accuracy on tall objects. They shaped the design: heavy data is stored per scene so the tool works offline, every disaster output is labeled a scenario, and a faster model must match the current one before replacing it. The reference data has limits too. Most laser tracks are years older or newer than the photograph, India has only two test sites per landscape, and laser labels can be wrong in dense towns.

**Table 5:** Known limits and what DepthWizard does about each today.

| Limit | Effect | What we do |
|---|---|---|
| Objects over 20 m are 15 to 18 m too low | Tall refuges look shorter | Warning on every result |
| Trees at 18 to 60% of true height | Forest heights unreliable | Tree-height map or laser correction |
| Ground under forest about 20 m off | Floods in forest unreliable | Flagged in the app |
| Mamba model less accurate so far | 7.20 against 3.90 m on aerial tiles | App keeps RS3DAda |
| Mamba model needs an NVIDIA graphics card | Not in the laptop version | Server use only |
| Only two Indian test sites per landscape | Wide confidence ranges | Ranges always reported |
| India's Cartosat images need a login | Not yet tested | Request to NRSC drafted |
| Some data is for non-commercial use only | Commercial use restricted | Research and government pilots |

## 10. Expertise Available with Each Student to Contribute to the Development of the Intervention

The project drew on machine learning, satellite data and web development, and just as much on research, writing, testing and presentation. Table 6 lists what each member brought to the team.

**Table 6:** Expertise of each student.

| Student | Expertise | Student | Expertise |
|---|---|---|---|
| Himanshu Patel | Full-stack web development and hosting, machine learning, satellite and map data | Saloni Mahajan | Writing, editing and figure layout |
| Shlok Vaidya | Model training, testing and benchmarks | Peruru Smitha Reddy | Presentation design and communication |
| Agampreet Singh | Research on floods and landslides in India | Akshita | Documentation and user guides |
| Saaransh Garg | Software testing and user feedback | Aryan Goyal | Data handling and literature review |
| Takshita Nagrale | Technical writing and editing | Arahana Garg | Human geography and disaster management |

## 11. Expected Outcomes

The main outcome is an offline tool that gives a checked height map from one image. We tested it on places the model never saw in training: 150 aerial tiles of three US cities with aircraft laser heights from the GAMUS set [20], 300 satellite tiles each of Atlanta and Buenos Aires from Overhead Geopose [21], and eight Indian sites checked against NASA lasers. Each number is an average with a 95% confidence range in brackets, set against a simple guess at the same points.

### 11.1 Height accuracy

**Table 7:** Average height error in metres with RS3DAda (lower is better).

| Test | Size | DepthWizard | Simple guess |
|---|---|---|---|
| Aerial tiles, every point | 150 tiles | 3.90 [3.24, 4.66] | 5.65 (flat ground) |
| Aerial tiles, buildings | 1,450 | 3.57 [3.08, 4.06] | 10.54 (flat ground) |
| Atlanta, satellite | 300 tiles | 5.99 [5.57, 6.43] | 5.67 (published) |
| Buenos Aires, satellite | 300 tiles | 1.94 [1.86, 2.03] | 2.01 (published) |
| India, against ICESat-2 | 8 sites | 4.85 [3.04, 6.69] | 8.25 (height 0) |
| India, against GEDI | 7 sites | 9.93 [4.86, 15.42] | 16.89 (height 0) |

Table 7 shows that for buildings, DepthWizard is about three times closer to the truth than guessing flat ground. At Indian sites it is almost twice as close as guessing zero. On satellite images it matches the results the original authors published. Figure 5 shows where the gain comes from: buildings and trees. On roads and bare ground the true height is near zero, so the flat guess is already right. Reading each tile four times instead of once lowers the error by 0.064 m on average and helps on 114 of the 150 tiles, so we kept it.

### 11.2 The Mamba model and other models

We ran four models on the same 750 tiles: RS3DAda, our Mamba student, and two versions of Depth Anything V2 [7]. Depth Anything does not give heights in metres, so we also score every

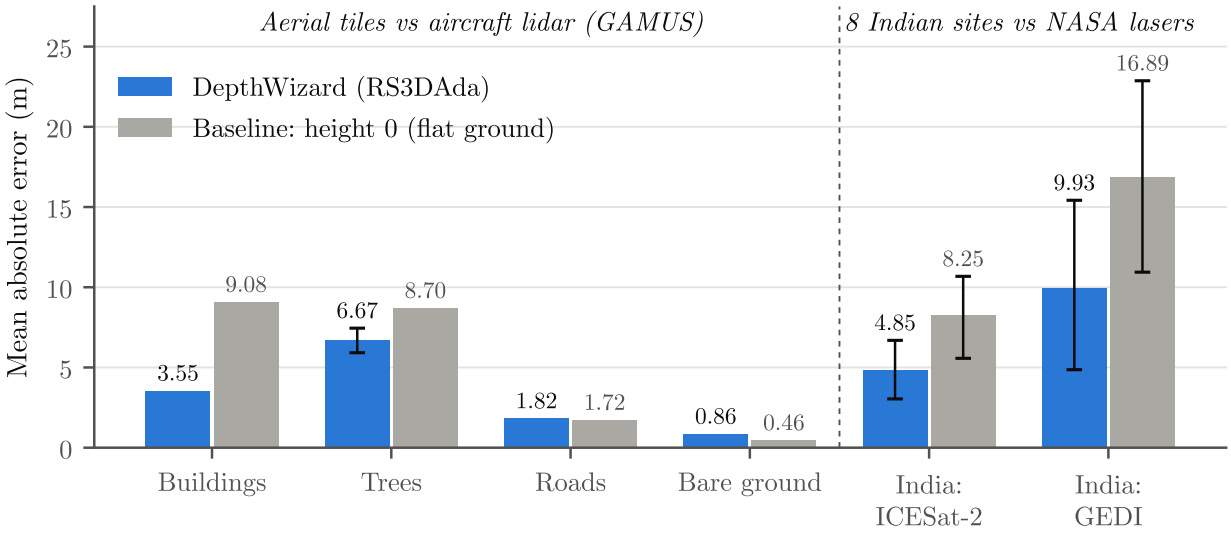



**Figure 5:** Average height error by type of surface. Left: aerial tiles against aircraft lasers. Right: Indian sites against NASA lasers, with 95% confidence ranges.

model after fitting its output to the truth with the best scale and offset for each tile; Table 8 calls this score "fitted". It shows how well a model finds where the tall things are.

**Table 8:** Height models on the same 750 tiles: average error in metres, and cost on one graphics card.

| | As predicted | | | Fitted | | |
|---|---|---|---|---|---|---|
| Model | Aerial | Atlanta | B. Aires | Aerial | Atlanta | B. Aires |
| Flat ground | 5.65 | 8.98 | 3.86 | | | |
| RS3DAda | **3.90** | **5.91** | **1.86** | **2.95** | **4.94** | **1.79** |
| Mamba student (ours) | 7.20 | 8.70 | 3.39 | 3.97 | 7.10 | 2.54 |
| Depth Anything V2, relative | | | | 3.25 | 5.64 | 2.18 |
| Depth Anything V2, metric | | | | 3.49 | 6.21 | 2.40 |

| | Size | Memory | Speed |
|---|---|---|---|
| RS3DAda | 366.5 M | 705 MiB | 105 tiles per minute |
| Mamba student (ours) | 40.7 M | 78 MiB | 546 tiles per minute |
| Depth Anything V2, relative | 335.3 M | 1,532 MiB | 118 tiles per minute |
| Depth Anything V2, metric | 335.3 M | 650 MiB | 134 tiles per minute |

The Mamba student is by far the cheapest model: nine times smaller than RS3DAda, with nine times less memory and 5.2 times the speed. Its first version is less accurate, 7.20 m against 3.90 m on the aerial tiles, where it loses even to the flat guess, and 8.70 m and 3.39 m against 5.91 m and 1.86 m on the satellite tiles. Even after fitting, its heights correlate with the truth at 0.29 against RS3DAda's 0.61, so it still struggles to find where tall things are. The app therefore keeps RS3DAda. RS3DAda also beats Depth Anything V2 on all three sets, by small but clear margins of 0.30, 0.70 and 0.39 m after fitting. Figure 6 shows typical tiles.

### 11.3 Tall objects and error ranges

Low objects are about right, but tall ones come out far too low, as Figure 7a shows. Above 20 m, heights are 16.26 m too low on aerial tiles and 18.31 m too low in India. The two satellite cities show the same thing, at 16.80 m and 15.25 m too low. This is the main weakness of the published model, and our Mamba student inherits it. The 90% error range held for 86.6%, 83.3% and 97.0% of

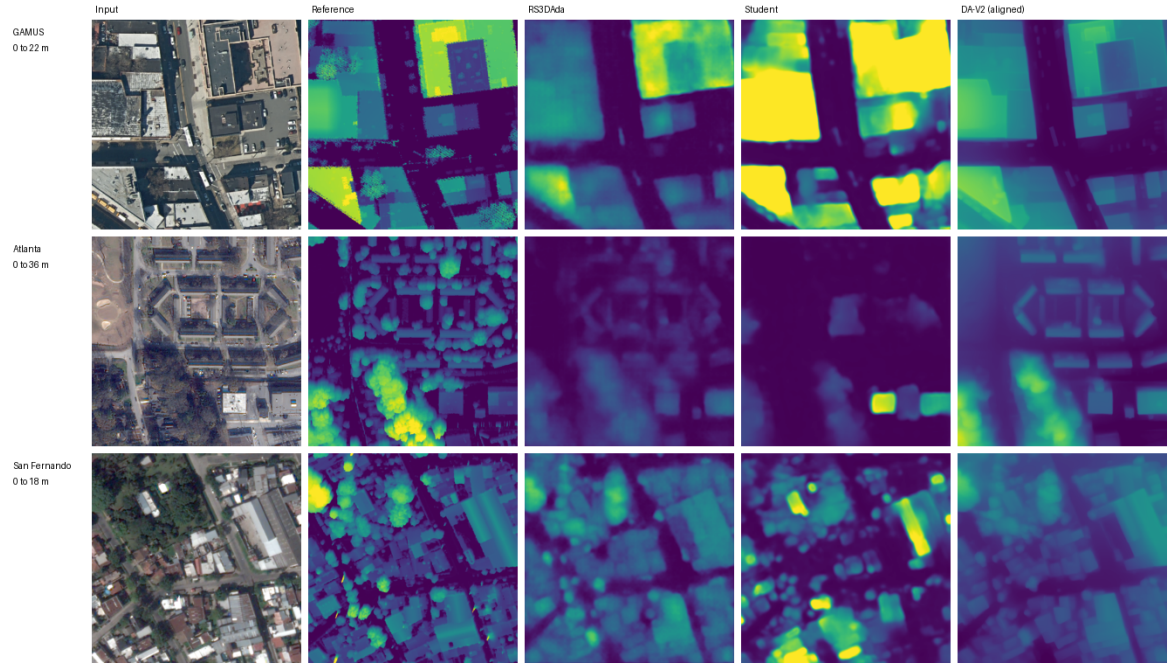



**Figure 6:** Typical tiles: input, laser truth, RS3DAda, our Mamba student and Depth Anything V2 (fitted). Rows: aerial set, Atlanta, Buenos Aires set. Dark is low and yellow is tall. The student finds buildings but makes them too tall on the aerial tile and blurs small objects.

building blocks in three cities it was not set on, with a width of about 6 m either side. On 96 Indian roofs it held for 71%. Figure 7b shows that it fails for buildings over 20 m, and the app warns users about this.

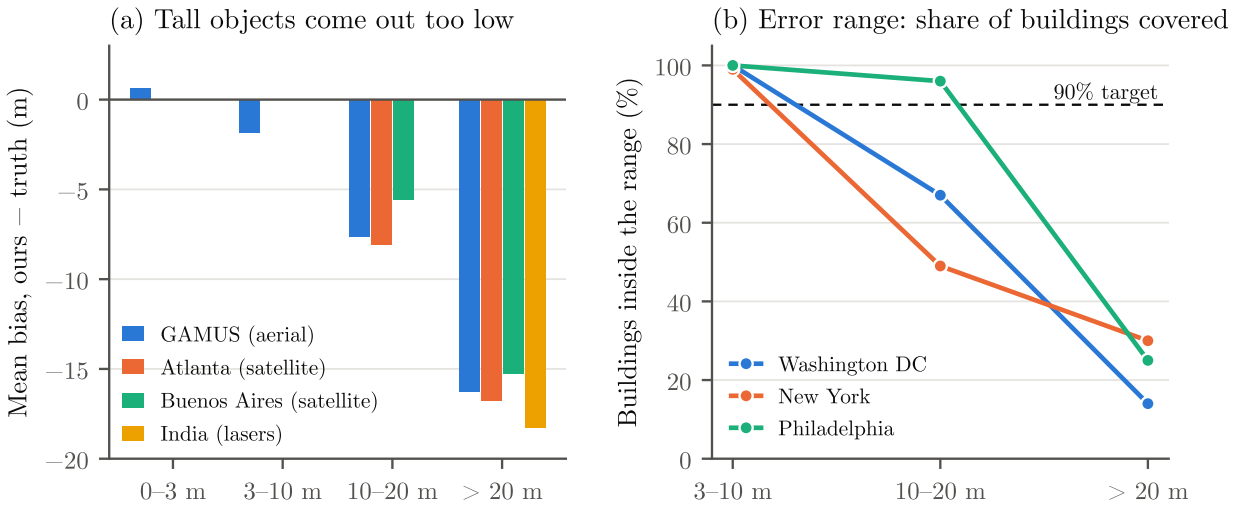



**Figure 7:** (a) Average error by true object height. (b) Share of buildings whose true height falls inside the 90% range, by height.

### 11.4 Ground level and floods

Removing rooftops and trees halves the ground map's error in Indian towns, from 5.28 m to 2.36 m, as Figure 8a shows. In hills FABDEM is better, and under forest every map is about 20 m off. Laser leveling removes the map's average offset, from 2.58 m too high to 0.27 m too low. Against GEDI's tree and roof tops our full surface beats the 30 m map, 5.97 m against 8.39 m, though not against all laser points, 9.14 m against 8.06 m; the photograph adds detail more than average accuracy.

The offset matters most for floods. We flooded the pre-flood heights of four areas of Porto Alegre, Brazil, and compared them with the mapped flood of 6 May 2024. Figure 8b plots the CSI: 0.20 with the free map, 0.30 with our cleaned map, 0.32 with FABDEM and 0.79 after laser levelling. The areas are small, so this tests the method rather than forecasting.

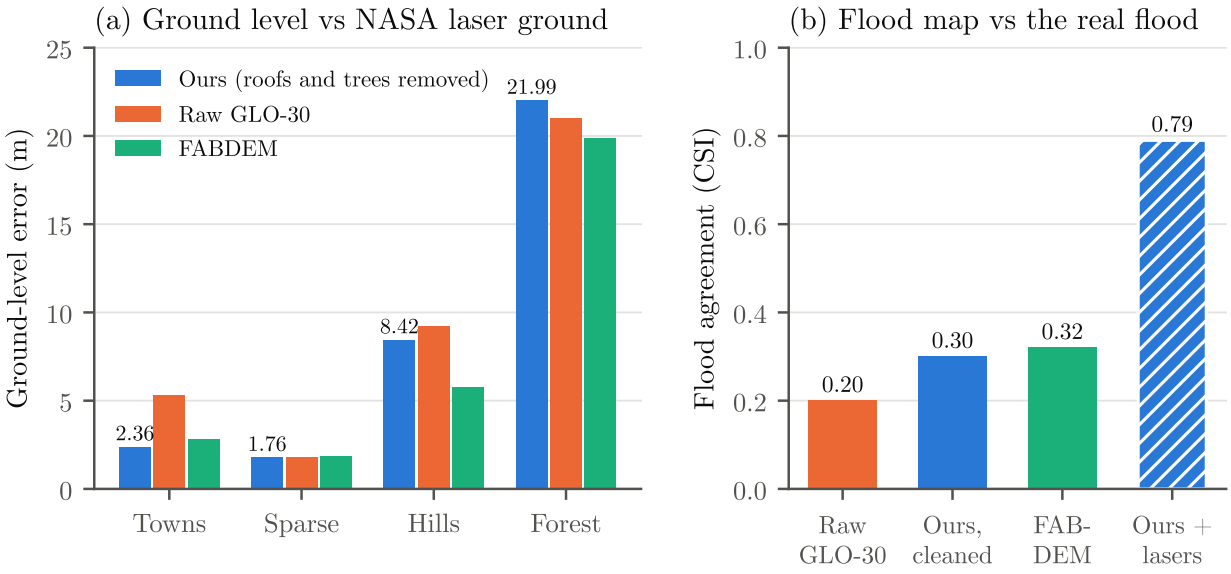



**Figure 8:** (a) Ground-level error against NASA laser ground at the Indian sites, by landscape. (b) Match between predicted and real flood maps in Porto Alegre with four ground maps. Striped: our map levelled with lasers.

### 11.5 A real Indian disaster: the Teesta flood, Sikkim, 4 October 2023

On 4 October 2023 the South Lhonak glacial lake burst and sent a flood down the Teesta valley. Using free public data only, we compared Sentinel-2 [23] satellite images from November 2022 and November 2023, the first cloud-free pair after the flood. Ground near the river that lost its vegetation or turned into water counts as damaged. We found 1.281 km$^2$ of damaged ground along 12 km of valley, mapped in Figure 9. The lake itself shrank by only 5.7%, from 1.358 to 1.28 km$^2$, but its water became much muddier.

We then ran DepthWizard on a pre-flood photograph of the valley [22] and loaded the damage map into the Observed-flood tool, shown in Figure 3f. The best-fitting flood scenario matches the damage map with a CSI of 0.49. It covers 86% of the damaged ground, but 47% of the ground it floods was not damaged. A single water level is a rough model for a fast flood in a steep valley, which explains much of the over-flooding. The damaged area reached 52 buildings, home to about 673 people by the app's estimate. Two cautions apply. The first clear images came a month after the flood, so the map shows what the flood left behind and not its peak. And each satellite pixel covers 10 m, which suits areas but not single houses.

### 11.6 What we tried that did not help

Some ideas were tested on the same data and dropped. A correction curve for heights helped in two cities and hurt in the third, where the error rose from 1.86 m to 2.87 m. Resizing the input image made no clear difference. Using GEDI points to make the laser levelling denser was slightly worse than using ICESat-2 alone. An error range for every pixel, in place of every building, held between 81.4% and 96.1% of the time across cities and missed its 90% target, so the app shows ranges for buildings only.

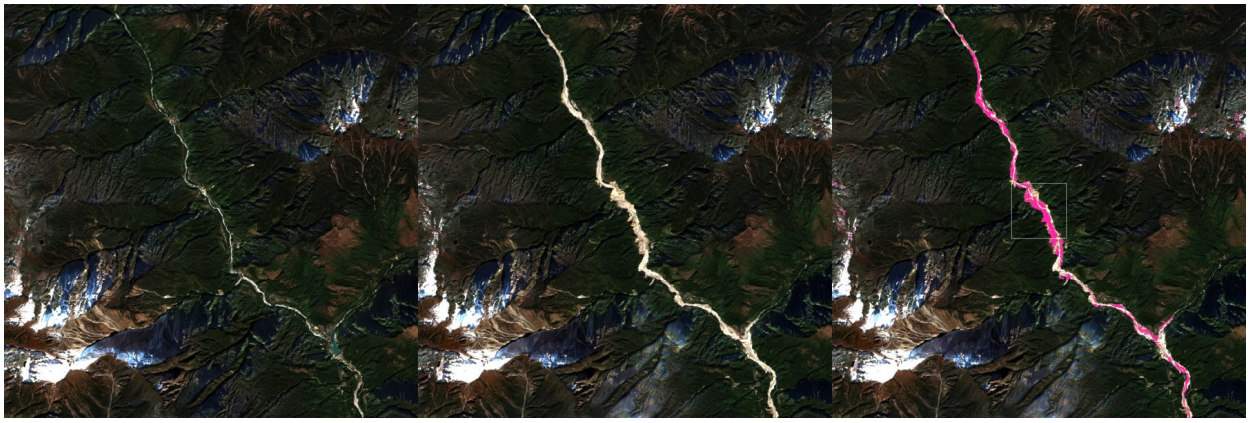


**Figure 9:** The Teesta valley before (left) and after (middle) the flood, and the damaged ground in magenta (right). White box: the area loaded into DepthWizard.

## 12. Suggested Plan of Action for Utilization of the Expected Outcome from the Work

The plan runs in four phases over about two years. Each phase starts only when the one before has passed its check.

**Table 9:** Plan of action, with the check that ends each phase.

| Phase and time | What we will do | Check before moving on |
|---|---|---|
| 1. Fix the model<br>0 to 6 months | Train the Mamba model for longer, add true heights from the synthetic cities, and adapt it to real images. Correct tall buildings from shadow length and sun angle. Measure the tree-height corrections on Indian forest sites. | Mamba error within 0.5 m of RS3DAda on the 750 test tiles; error above 20 m cut by half |
| 2. Test on Indian data<br>3 to 9 months | Get Cartosat images and CartoDEM from NRSC and repeat every test. Add at least ten sites for each landscape, with ground points from a state survey department. | Results hold on Indian imagery; confidence ranges narrow enough to compare landscapes |
| 3. Pilot with a state<br>one monsoon | Work with one State Disaster Management Authority, for example Assam or Himachal Pradesh. Before the rains, give flood-reach and landslide maps for two or three districts. After each flood, compare them with the mapped extent and with field reports. | District officers use the maps in their monsoon plans; flood match (CSI) of 0.6 or more on real events |
| 4. Hand over and scale<br>from year 2 | Give the code, test register and user guide to NRSC and ISRO so the model can run on their servers and feed Bhuvan. Train district staff with short hands-on sessions, and offer the tool to geography and planning courses. | A state or national agency runs it without our help |

The same height maps can also serve town planning, rooftop solar studies and checks on unauthorised construction, at any phase.

## 13. Conclusion

One satellite photograph, with free ground data and NASA's lasers, gives a 3D height map that disaster teams can use offline and check. DepthWizard beats simple guesses on every height test, halves the ground error in Indian towns, and maps a real Indian flood from public data. Our Mamba model is nine times smaller and five times faster; making it as accurate is the next job. Very tall buildings and forests remain hard, and every result carries its error so users know where to trust it.

### Data and code

The code, the register of every number, a video walkthrough and the steps to run the web demo are in the project repository: https://github.com/hima1323/terravista-depthwizard. All test data are public.

## 14. Contribution of Each Student of the Group to Complete the Assignment

**Table 10:** Contribution of each student to the project and this report.

| Student | Contribution |
|---|---|
| Himanshu Patel | Led the project. Built the whole website and app, front end, back end and hosting; the ground-level correction; the Mamba model code; the eighteen tools; the Teesta case study; and wrote this report |
| Shlok Vaidya | Built checking in the app and survey-point correction, the installer, and the helipad and route tools; trained the Mamba model and ran the model comparison |
| Agampreet Singh | Gathered reports on past floods and landslides in India for the problem definition |
| Saaransh Garg | Tested the demo on the sample scene and reported interface problems |
| Takshita Nagrale | Edited the report and checked it against the HS202 format |
| Saloni Mahajan | Helped select and arrange the figures and screenshots |
| Peruru Smitha Reddy | Designed the presentation slides |
| Akshita | Reviewed the README and the demo instructions |
| Aryan Goyal | Checked the numbers in the result tables against the source files |
| Arahana Garg | Summarised disaster-management guidance used in the background section |

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